

complex for efficient control of *Salmonella* in broiler chickens. The study is in progress to evaluate the vaccine potential of this complex using HD 11 activation assay and chicken challenge study.

508P Evaluation of the effect of *Histomonas meleagridis* morphology on resiliency and survivability when subjected to incubation at room temperature. Alessandro Rocchi^{*1}, Aaron Forga¹, Billy M. Hargis¹, Danielle Graham¹; ¹University of Arkansas, Poultry Science, Fayetteville, Arkansas, United States.

Histomonas meleagridis, a protozoal parasite of gallinaceous birds, is the etiological agent for histomonosis, or blackhead in turkeys. *H. meleagridis* has been shown to have poor survivability, surviving only a few hours, outside of a host except when within eggs of the cecal nematode *Heterakis gallinarum*. The purpose of the present study was to evaluate the survivability and morphological changes of *H. meleagridis* in 25cm² tissue culture (TC) flasks at room temperature (RT-21°C) as compared to TC flasks incubated at 40°C (INC-40°C). Survivability was evaluated in separate tests for two morphologies of *H. meleagridis* upon reaching confluence: 1) amoeboid forms expressing movement (Am test) or 2) spherical forms expressing no movement (Sp test). Histomonads were grown to confluence and desired morphology in Modified Dwyer's media (MDM). For the assay, 9mL of confluent culture was transferred into 25cm² RT flasks (n=3) and INC flasks (n=3). Cultures were monitored for changes in morphology, concentration, and activity. Histomonads were enumerated pre-transfer (d0) and d3 and d7 post-transfer. Bacteria within the cultures were enumerated via 10-fold serial dilution drop plating on Chromogenic Orientation agar at d0, d3, and d7. *Escherichia coli* and *Klebsiella pneumoniae* were recovered and identity confirmed by MALDI-TOF. ANOVA was used to determine significant differences at P<0.05. At d7 post-transfer, 1ml/flask was passaged into 25cm² TC flasks with fresh MDM media to evaluate viability. Histomonad concentrations (Log₁₀ histomonads/ml) for Am and Sp were 5.58 and 5.42 at d0, respectively. Am-RT flasks experienced a shift in concentration from d1-d2 when the number of histomonads decreased, becoming nearly undetectable at 40x magnification. Sp-RT flasks experienced a decrease in concentration at d5. Sp-INC showed no concentration changes at d5 although histomonads were visibly smaller compared to d0. No other visible changes were seen in the RT flasks for either the Am or Sp test. Only the INC flasks were able to reach confluence after d7. There were no statistical differences (P>0.05) in bacterial recovery at all timepoints evaluated indicating that RT and INC conditions did not negatively impact viability. Overall, it appeared that after experiencing the major shifts seen on d2 for the Am flasks and d5 for the Sp flasks the histomonads were present although not viable for culturing. The starting morphology of the histomonads (Am vs. Sp) when subjected to RT also appeared to influence the longevity and time before reaching any major morphological shift.

509P Effects of alkali-hydrolyzed corn dried, distiller grains with solubles on growth performance and meat quality of broilers under heat stress. Euyeon Noh^{*1}, Byungrok R. Min¹, Amit Kumar Singh², Woo K. Kim²; ¹University of Maryland Eastern Shore, Princess

Anne, Maryland, United States; ²University of Georgia, Athens, Georgia, United States.

The misuse of antimicrobial growth promoters (AGPs) in animal production has contributed to antimicrobial resistance. Phytochemicals such as phenolic acids are considered promising alternatives to AGPs. Corn dried distiller grains with solubles (DDGS) contain considerable amounts of phenolic acids, but their bioavailability is extremely low. In our previous study, alkali-hydrolyzed DDGS (AH-DDGS) was developed as a source of bioavailable phenolic acids, promising alternatives to AGPs. This study aimed to determine the effects of AH-DDGS as an alternative AGP and remedy against heat stress on growth performance and meat quality of broilers. A total of 640 one-day-old broiler male chicks were randomly assigned to 32 floor pens with 20 birds/pen (8 pens/treatment). They were raised for 42 days and fed corn/soybean meal diets with 4 treatments: 1) negative control (NC), commercial DDGS (5%), 2) positive control (PC), commercial DDGS (5%) and AGP (Bacitracin methylene disalicylate 50 (500g/tonne of feed)), 3) T1, AH-DDGS (5%, prepared by alkali hydrolysis using Ca(OH)₂ at 0.45 mol/kg wet distillers grains), and 4) T2, commercial DDGS (2.5%) and AH-DDGS (2.5%). Heat stress was applied to all treatments for 8 h/day during the grower (d15-28) and finisher (d29-42) periods in which the temperature was increased to 35 °C and 33 °C, respectively. At d43, 5 birds/pen were slaughtered and breast meat was harvested after 24 h postmortem. Growth performance of broilers, post-chill carcass yield, and breast meat quality parameters were determined. Data were analyzed using one-way ANOVA with Duncan's multiple range test for mean comparison. T1- and T2-fed broilers showed similar growth performance to PC-fed which showed higher growth performance than NC-fed (P<0.05). Carcass yields were not different among the treatments. Marinate uptake was significantly higher in the breast meat from T1-fed broilers than those from others (P<0.05). When cooked at 72 h postmortem and stored for 7 days at 4 °C, breast meat from T1- and T2-fed broilers showed significantly higher cooking yields and lower lipid oxidation rates than the controls (P<0.05). Other quality parameters such as color, drip loss, and textural properties were not different among the treatments. The results showed that AH-DDGS provided similar effects on the growth performance of broilers to AGPs and improved oxidative stability of broiler breast meat, leading to partially improved muscle protein functionalities. Hence, the results indicated that AH-DDGS rich in bioavailable phenolic acids has high potential as an economically feasible alternative to AGPs and remedy against heat stress to maintain/improve broiler health and productivity and their meat quality.

510P A specialized hand-held trimming device for removing wishbones and cartilages from poultry breast meat. Michael Park^{*1}, Sean L. Thomas¹, Aklilu Giorges¹; ¹Georgia Tech Research Institute, Atlanta, Georgia, United States.

Trimming unwanted bones, fats, and gristle from poultry breast meat after deboning remains a highly manual task. The removal of wishbones and cartilages from the butterfly-shaped breast meat cut is one of the most frequent and challenging components of breast trimming. Currently, a blade knife is predominantly used for the task, resulting in

high yield loss and slower throughput. The overall objective of this study is to design a specialized trimming tool and method to remove cartilages and wishbones from breast meat, aiming to increase yield, throughput, and worker comfort. The trimming tool is a hand-held device that consists of a cylindrical blade attached to a pneumatic spinning motor. Although the system is capable of reaching 25,000 RPM, a motor with 10,000 RPM was found to be sufficient. Furthermore, an approach angle of 22.5° achieved the most efficient cut, and the device handle was tapered to minimize the bending of the user's wrist for added comfort. The trimming time and yield loss of the new rotating tool were compared to those of a blade knife in order to assess its performance. Twelve breast butterflies were harvested from the front halves, with wishbones and cartilages fully intact. The time taken to remove each wishbone and cartilage from each sample was recorded, and the weight of the removed bones and cartilages was measured. The yield loss was estimated by calculating the difference in mass between the initial and remaining bones and cartilages after scrapping. The average trimming time for wishbones and cartilages using the rotating tool was 9.21s and 9.51s, respectively, compared to 15.7s and 16.2s for the blade knife. Additionally, the standard deviation of the trimming time for the rotating tool was 2.40s and 1.17s, while for the blade knife it was 5.80s and 3.58s, respectively. For the average weight of the lost meat on wishbones and cartilages, the rotating tool resulted in 4.88g and 5.37g, with standard deviations of 2.25g and 1.51 g, respectively. In comparison, the blade knife had an average weight of 5.60g and 7.93g, with standard deviations of 1.50g and 2.51 g, respectively. To determine the statistical difference in time and mass losses between the two trimming methods, a student t-test was conducted. The results showed that the trimming time for bones and cartilages was significantly shorter when using the rotating tool. As for mass losses, the new method reduced losses from cartilage trimming, while there was no significant difference in mass losses between the two methods for wishbone.

511P Temporal profiling of the Bursa Transcriptome reveals systemic differences induced by Marek's Disease virus. Oladipupo Bello^{*1}, Qin Chu², Jlu Zhou Song³; ¹University of Maryland, Animal Sciences, College Park, Maryland, United States; ²Beijing Academy of Agriculture and Forestry Services, Institute of Animal Husbandry and Veterinary Medicine, Beijing, China; ³University Maryland, Animal and Avian Sciences, College Park, Maryland, United States.

Marek's disease (MD) is an acute lymphoma-inducing disease in chickens caused by Marek's disease virus (MDV). The study aimed to identify pathways, biological processes, genes, and temporal variations that might be critical to MD resistance by examining transcriptional changes in the bursa of two highly inbred White Leghorn chicken lines (Line 63 - MD-resistant; Line 72 - MD-susceptible) in response to MDV. Experimental design is a completely randomized design with 3 factors: Line, Time, and Status. Fourteen days after hatching, 8 unsexed chickens from each line were inoculated with 500 PFU of a very virulent plus strain of MDV. Another set of 8 unsexed chickens from each line served as age-matched controls. In each line, 2 infected and 2 control chickens were terminated at 5, 10, and 21 days post-infection (DPI), corresponding to the early cytolytic, latency, and transformation phases of the disease, respectively. Bursa tissues were extracted from the carcasses, followed by RNA isolation and sequencing. Differential gene expression analyses comprising within-line (infected vs. control) and between-line (infected vs. infected) contrasts at each time point were performed using likelihood ratio test (FDR = 0.05) implemented in edgeR. GO term and pathway enrichment analyses of the differentially expressed genes (DEGs) were performed using hypergeometric test in gprofiler2 (FDR = 0.05). There was enrichment of proline metabolism pathway at 5 DPI with DEGs upregulated in the resistant birds compared to the susceptible birds, especially with P4HA2 gene, which has an expression level four times in the resistant birds. Previous research has shown the significant role of proline metabolism in cell signaling, including immune cell signaling. At 10 DPI, there was significant enrichment of T cell differentiation, lymphocyte co-stimulation, and Cytokine-cytokine receptor interaction pathway with DEGs upregulated in the resistant birds compared to the susceptible birds. The effect of P4HA2 gene at the early cytolytic phase might play a role in the better response mounted against the infection in the resistant chickens at 10 DPI. Even though there was a significant enrichment of immune-related biological processes and pathways, including cytokine-cytokine receptor interaction pathway, with DEGs upregulated in the susceptible chicken compared to the resistant birds at 21 DPI, the better response by the resistant birds at 10 DPI—the only stage devoid of viral replication—is a well-timed response. In summary, pathways and processes that are likely critical in host response to MD were identified. Likewise, the results showed the timing of response as a plausible explanation for the differential resistance to the virus.